Electronics – EE3011 Fall 2021-2022

Course Description

This course introduces basic concepts of semiconductors and teaches common diode and transistor circuits. The course aims to develop a basic understanding of the theory and practice of electronic devices. Students who complete the course will gain the skills necessary to analyze and design simple electronic circuits.

Date and Time

Lectures will be held **in RTEM4.Z01 on Tuesdays from 10:30 to 12:20** and will be held **in RTEM1.119 on Thursdays from 13:00 to 13:50**.

If needed due to the Covid19 restriction, we will have online lectures:

https://zoom.us/j/98575852185?pwd=cU9vaTJkMXVidkU2RGtoV1BmdTFRZz09

Meeting ID: 985 7585 2185, Passcode: 159613

Instructor & Assistant

The course will be given by Dr. Ahmet Unutulmaz.

E-mail: ahmet.unutulmaz@marmara.edu.tr, Room: M4-249

The course will be assisted by Fethullah Sipahioğlu (fsipahioglu@marmara.edu.tr), Salih Çolakoğlu (salih.colakoglu@marmara.edu.tr) and Kemal Bayat (kemal.bayat@marmara.edu.tr).

Office Hours

Students are welcomed to discuss the course materials on Tuesdays from 12:20 to 13:00 in the instructor's office (M4-249).

Textbook

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet **Microelectronic Circuits** 8th Edition, *Oxford University Press*, 2019. ISBN: 9780190853464

Homework

During the semester 5 homework are going to be assigned. Each will contribute 3 points to your overall grade. Sadly, **any late submission will not be accepted.**

Lab Sessions

There are going to be eight labs. Some of the labs will be held online and some of them will be face-to-face **in RTEM4.101**. Each lab will contribute 3 points to your overall grade. **Any student who misses more than 2 labs will fail.** There will be 4 lab sessions, students must join the one they registered for.

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Meeting ID: 985 7585 2185, Passcode: 159613

Lab Session	Day and Time
1	Thursdays from 08:30 to 10:20
2	Thursdays from 14:00 to 15:50
3	Thursdays from 16:00 to 18:00
4	Fridays from 17:00 to 18:50

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Grading Policy

Homework 15%, Laboratory Grade 24%, Midterm Exam 20%, Final Exam 40%, Bonus 1%.

Midterm and Final Exams will be face-to-face, and their grades will count for 20% and 40% of the semester grade, respectively.

Midterm exam will be given in week 8 between November 20th and November 28th. And the final will be given between January 24th and February 6th. Exact dates and times will be announced later.

Letter grades will be assigned according to DDS: 0-44 FF, 45-49 FD, 50-54 DD, 55-64 DC, 65-74 CC, 75-79 CB, 80-84 BB, 85-89 BA, 90-100 AA.

Make-up Policy

Student, whose excuse is approved by the University, will be given a make-up exam.

Lab Schedule

Lab	Date	Subject
1*	November 4th & 5th	Introduction to LTSpice
2*	November 11th & 12th	Simulation of Diode Circuits
3	November 18th & 19th	Diode Characteristics
4	December 2nd & 3rd	Diode Rectifier with Peak Detector
5*	December 9th &10th	MOSFET Characteristics
6	December 16th & 17th	BJT Characteristics
7	December 23rd & 24th	Biasing BJT Circuits
8	January 13th & 14th	The Common Emitter Amplifier

*Labs 1,2 and 5 will held online.

Course Schedule

Week	Date*	Subject
1	October 5th & 7th	Introduction
2	October 12th & 14th	PN Junction
3	October 19th & 1st	Diodes
4	October 26th	Diodes
5	November 2nd & 4th	Diodes & MOSFET
6	November 9th & 11th	MOS Field Effect Transistors (MOSFET)
7	November 16th & 18th	Biasing MOSFETs
8	-	Midterm Week
9	November 30th & December 2nd	Bipolar Junction Transistors (BJT)
10	December 7th & 9th	BJT
11	December 14th & 16th	Biasing BJTs
12	December 21st & 23th	Small-Signal Operation
13	December 28th & 30h	Small-Signal Models
14	January 4th & 6th	Common Source/Emitter Configuration
15	January 11th & 13th	Common Gate/Base Conf. and Source/Emitter Follower
16	January 18th & 20th	Discreate Circuit Amplifiers

*There will not be a lecture on October 28th.